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Barriers to and facilitators of physical activity among community-dwelling older adults: a systematic review

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ABSTRACT

Objective This study aims to systematically review the evidence on barriers to and facilitators of physical activity (PA) engagement among community-dwelling older adults.

Design Systematic review.

Data sources Electronic databases MEDLINE, Embase, APA PsycINFO, SPORTDiscus, CINAHL Plus, AgeLine and Scopus were searched from their inception to 14 April 2024.

Eligibility criteria Eligible studies were qualitative, quantitative or mixed-methods studies, published in English and investigating barriers to and/or facilitators of PA among independently living adults aged ≥65 years.

Data extraction and synthesis Content analysis was performed using a narrative synthesis adapted from the socioecological model of health behaviour by three independent assessors. Barriers and facilitators were categorised into intrapersonal, interpersonal and environmental factors.

Results Of 27 779 screened articles, 20 eligible studies were synthesised (11 qualitative, 1 mixed-method and 8 quantitative), with a total of 18 597 participants. The main barriers to PA included intrapersonal factors: fear of injury, pain, low motivation and lack of spare time; interpersonal factors: lack of social support and family obligations; environmental factors: poor weather, poor built environment/high cost accessing PA facilities. The main facilitators to PA included intrapersonal: health and well-being, fitness devices, better self-perceived body image, sense of bodily control and enjoyment; interpersonal: social support and having a partner/group/pet to engage in PA together; environmental: age-friendly built environment and access to PA programmes.

Conclusion There are multiple barriers to and facilitators of PA engagement among community-dwelling older adults that should be considered by policy makers when designing PA programmes to help increase PA in older adults.

PROSPERO registration number CRD42023417404.

STRENGTHS AND LIMITATIONS OF THIS STUDY

⇒ This study reviewed qualitative, quantitative and mixed-method studies exploring the barriers to and facilitators of physical activity (PA) engagement among community-dwelling adults aged 65 years and above.
⇒ Rigorous synthesis of data was conducted using the socioecological model of health behaviour to frame themes and subthemes.
⇒ Included studies were mostly conducted in high-income countries and the participants were mostly Caucasian, limiting generalisability.
⇒ Studies were limited to those published in English language.
⇒ The barriers to and facilitators of PA identified within our study population may be very different to older adults living in care facilities, who may be living with a functional disability or a life-limiting condition.

INTRODUCTION

The global population is ageing, with a projected twofold increase in the number of individuals aged 65 years and older; from 703 million in 2019 to 1.5 billion by 2050.[1] Although the global ageing population is increasing, not all older adults are able to age healthily and maintain their functional independence throughout their lifetime.[1] It is estimated in 2014 that in some countries, such as Slovakia, only 16% of women and 23% of men aged 65 years and older are able to age healthily.[2] Additionally, while 73% of women and 77% of men aged 65 and older in countries like Sweden are able to age healthily, there is still potential for improvement.[2]

Physical activity (PA) is defined as 'any bodily movement produced by skeletal muscles that requires energy expenditure, such as engaging in exercise, sports, activity of daily living, leisure time movements and active transportation'.[3] Engaging in regular PA can help prevent and control non-communicable diseases including cardiovascular disease,[4] certain cancers,[5] diabetes,[5] improve mental health[6] and delay the onset of dementia.[7] Engaging in regular PA can also assist older adults in retaining their functional capacity for a longer period and delaying the need for assistance with activities of daily living.[8]

The WHO recommends that older adults (aged ≥65 years) engage in at least

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150–300 min of moderate or 75–150 min of vigorous PA per week, at least 2 days of muscle strengthening activities and a multicomponent exercise on at least 3 days a week.[3] Duration (how long), frequency (how often), intensity (light/moderate/vigorous) and mode (walking, gardening, running, swimming) of PA will impact on an individual's functional ability, although at least 30 min of PA 5 days per week is recommended for older adults.[3] For adults with chronic or physically limiting diseases, such as muscular dystrophy or osteoarthritis, it is recommended they are active as their abilities and conditions allow.[3] However, most older adults are not engaging in PA at the WHO recommended levels.[9] A 2024 cross-sectional study of 5.7 million adults surveyed using the International Physical Activity Questionnaire from 197 countries and territories identified that 31.3% of respondents were found to be engaging in PA below the recommended levels.[9] While the lower than recommended engagement in PA among the older adult population may be partially attributable to chronic diseases and the effects of ageing, these may not be the only factors limiting older adults' engagement in PA.

Identifying and understanding the barriers to and facilitators of engaging in PA among older adults is essential to inform future health policies and programmes aimed at increasing PA participation in this growing population. With the number of adults aged 65 and over projected to double by 2050, promoting healthy ageing is critical.[1] Encouraging PA engagement can help older adults maintain functional independence and continue performing activities of daily living as they transition from middle to older age.

Several systematic reviews have synthesised the barriers to and facilitators of PA engagement among older adults participating in a PA intervention or programme,[10 11] living with a life-limiting disease,[12] living in a residential care facility[13] or in specific population groups such as retirees of lower occupational groups.[14] Other studies have only investigated the qualitative synthesis of barriers to and facilitators of predominantly structured PA,[15] or among populations aged 80 already living with frailty or having some type of chronic comorbidity.[16] Evidence is limited on the barriers to and facilitators of PA engagement among community-dwelling older adults who do not have functional disabilities and are not receiving palliative care. One existing review used the Andersen Model of Health Service Utilisation to summarise such barriers to and facilitators among older adults without functional disability or in palliative care.[17] However, this model does not fully explain why older adults engage or fail to engage in PA for recreation, transportation or leisure within community settings. It also does not adequately account for the dynamic interactions between individuals and their environments that influence health and well-being. Our study seeks to extend the available evidence base by systematically reviewing and synthesising the literature on the barriers to and facilitators of PA engagement among community-dwelling older adults. We adopt the socioecological model of health behaviour as our guiding framework and include quantitative, qualitative and mixed-method studies.

METHODS

This systematic review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) framework.[18] The review protocol was registered in PROSPERO (reference number: CRD42023417404).

Search strategy

The search strategy was conducted with guidance from a senior medical librarian. Electronic searches of MEDLINE, Embase, APA PsycINFO, SPORTDiscus, CINAHL Plus, AgeLine and Scopus databases were performed from their inception to 14 April 2024. No restrictions on the date of publication were placed; however, included articles were limited to 'English Language'. The search strategy included the following main search terms and concepts: "aged", "barrier", "facilitator" and "physical activity", and was tailored to each of the seven databases (Search strategies are available in online supplemental tables 3-5).

Study selection

Studies that met the following criteria were eligible for inclusion in the review: (1) qualitative, quantitative or mixed-method design, (2) included apparently healthy community-dwelling adults aged 65 years and older (baseline), (3) investigated barriers to and/or facilitators of PA, (4) focused on any mode, intensity or PA programme as defined by researchers, (5) were published in English language.

Studies were excluded if participants (1) had a functional disability at baseline (including functional decline or dementia) or were receiving palliative care, (2) had recently recovered from a disease or hospitalisation, (3) were undergoing hospitalisation or rehabilitation at study baseline, (4) were institutionalised (such as living in aged care facilities). Additionally, literature review, scoping review, systematic review, meta-analysis, narrative review, case study or anecdotal study types were excluded.

All relevant studies were screened on Covidence systematic review software.[19] Screening was performed by two independent investigators. After importing all the search results from the seven databases to Covidence, duplicate articles were automatically removed and the remainder were screened initially by title and abstract by two reviewers (YaC and SS). Articles meeting the selection criteria after final full-text review were included in the review. During the screening stage, any discrepancies were resolved through discussion by the two investigators (YaC and SS). A third investigator (DG) was consulted in case a resolution was not achieved. Missing full text or results was resolved by emailing the authors of the paper, requesting access.

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Data extraction

Data extracted consisted of details about the first author and year of publication, country/region, study design, study duration, participant characteristics (sample size, mean age and/or age range at baseline, ethnicity), data collection method, interview time (qualitative studies), follow-up period (quantitative/mixed-method studies, if applicable), PA definition and assessment, barriers and facilitators definition and assessment. The data extraction process was performed independently by two investigators (YaC and SS). Both investigators read and familiarised themselves with the articles included. Where one investigator (YaC) extracted all the data into a Microsoft Excel table, a second investigator (SS) checked the data extracted to detect any potential inconsistencies.

Data synthesis

A content analysis was performed to synthesise data.[20] A content analysis was primarily selected as this study investigates quantitative, qualitative and mixed methods studies, and a content analysis can help sort qualitative findings into themes and subthemes, which can then be compared with the quantitative findings more comprehensively.

Three authors (YaC, YuC and SS) completed the data synthesis. Data were synthesised into three main levels, using a coding framework, adapted from the socio-ecological model (SEM) of health behaviour: intrapersonal, interpersonal and environmental factors.[21] For each level, a priori themes were created to construct a codebook in a Microsoft Excel spreadsheet, based on previous literature,[10–17] which included: (1) intrapersonal (eg, physical, psychological and economic influences); (2) interpersonal (eg, social and cultural); (3) and environmental (eg, access to facilities, policies and the built environment). New themes and subthemes were developed inductively using a narrative synthesis, developed from the words, phrases or paragraphs identified through the included studies. Each author independently read the identified studies twice, extracting data which were categorised into one of the three levels of the SEM. Data were allocated to more than one theme if it satisfied the definition of more than one theme. The codes that emerged from the data were then compared, and any discrepancies were resolved by a fourth researcher (DG). To improve the transparency of researcher agreements, an inter-rater reliability score was calculated, with agreements reaching 82% across all three authors. Other theoretical models such as the stages of change model or self-efficacy model were not adopted, as environmental factors were less integrated into the theoretical concept.

All quantitative and mixed-methods results were reported using proportions (%). All qualitative data were reported descriptively.

Quality assessment

To minimise and prevent bias, qualitative studies were appraised by YaC and SS using the Critical Appraisal Skills Programme (CASP),[22] quantitative studies using a Joanna Briggs Institute (JBI) checklist for prevalence studies[23] and mixed-method studies using the Mixed Methods Appraisal tool (MMAT).[24] Qualitative studies were scored out of 10 using the CASP checklist, quantitative studies out of nine using JBI and mixed-method studies out of seven using MMAT. Included studies were rated against each of the checklist criteria and returned with one of three responses 'yes', 'no' or 'not clear'. A score of one was given if a 'yes' was returned. Studies with higher scores were considered to have a higher level of congruity between the study methodology and each checklist criterion (CASP for qualitative, JBI for quantitative and MMAT for mixed-methods studies), and thus deemed to be of higher quality. Studies classified as low quality may have flaws in the study's methodology and need to be interpreted with caution.[23]

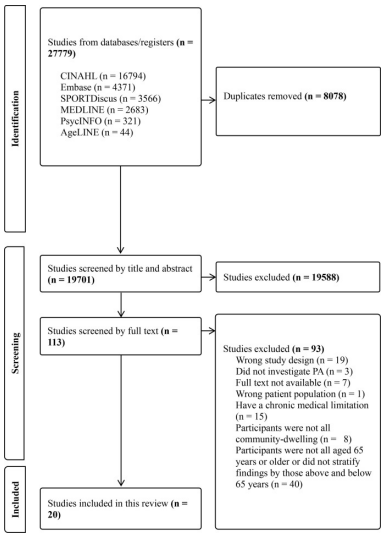


Figure 1 PRISMA flow diagram for systematic review of barriers and facilitators to engagement in physical activity among healthy community-dwelling adults aged 65 years and older. PA, physical activity; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

Patient and public involvement

None.

RESULTS

Study characteristics

The PRISMA flow diagram is shown in figure 1, providing the number of search results and included articles. 20 studies were included, consisting of qualitative (n=11),[25–35] mixed-method (n=1)[36] and quantitative (n=8) studies,[37–45] with a total of 18 597 participants. Most of the studies were conducted in the USA (n=6),[25 27 30 31 35 41] followed by UK (n=3),[26 29 36] Australia (n=3),[33 37 39] Germany (n=1),[40] Norway (n=1),[32] Canada (n=1),[34] Taiwan (n=1),[44] India (n=1),[42] Finland (n=1)[43] and multiple European countries (n=2).[28 38] Most studies did not report the ethnic backgrounds of the participants (n=12)[27 28 32 34–40 42 43] and for the studies that did, most

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participants were Caucasian[25 26 29–31 41] or Asian.[33 44] The number of participants in each study ranged from 9[32] to 8881.[35] Most studies had a greater proportion (≥50%) of female participants, except two studies where the proportion of female participants was less than 50%,[29 36] and one study included only men.[32] The age of participants ranged from 65 to 99 years.

Most qualitative and mixed-methods studies (n=7) collected data on barriers and facilitators using semistructured interviews,[26 27 29 30 33 34 36] while some (n=5) used focus groups.[26 28 31 32 35] Where reported, the length of the interview/focus group time varied from 14 min to 149 min.

All quantitative studies were cross-sectional by design, while an identified mixed-method study was a prospective cohort study with a follow-up period of 5 years. All quantitative data about barriers to and facilitators of PA engagement were collected via questionnaires or surveys (online supplemental tables 1,2). Results are summarised in themes and subthemes in table 1 and described below. A summary of qualitative and mixed-methods against quantitative and mixed-methods results is shown in online supplemental table 9.

To better highlight differences in outcome reporting, the results are also presented as a Venn diagram in figure 2, which visually depicts only those barriers, facilitators and factors that were reported by included studies as both barriers to and facilitators of PA. To clarify how a factor was described as acting both as a barrier and a facilitator, differences in wording are summarised in table 2.

Quality of evidence

Out of the 11 qualitative studies appraised using the CASP checklist, 4 studies received full scores, 3 scored 9 out of 10, 3 scored 8 out of 10 and 1 study scored 7 out of 10 (online supplemental table 6). The scores were primarily reduced due to the use of an unsuitable recruitment strategy for the study's objective and a lack of rigorous data collection methods. For the one mixed-methods study appraised using the MMAT checklist, six out of seven were received for the level of evidence (online supplemental table 7). Out of the eight quantitative studies appraised using the JBI checklist for prevalence studies, four studies received full scores, two scored eight out of nine and two scored seven out of nine. The scores were deducted due to uncertainties in how inadequate response rates were managed (online supplemental table 8).

Barriers to physical activity engagement

Intrapersonal

Psychological: The fear of falling/experiencing an injury was one of the most commonly reported barriers to engaging in PA.[25 27 30–35 39 40] Some participants highlighted specific types of fear during engagement in PA, such as fear of fracture/damage to joints,[26] getting hit by a car[34 43] or getting assaulted.[34] During the COVID-19 pandemic, some also described fear of contracting COVID-19 as a barrier to engaging in PA.[29] Some participants reported that they were afraid of developing an injury and would not conduct any type of PA unless they received reassurance beforehand from their general practitioner, osteopath or a physiotherapist.[26 43]

Health-related issues: Among older adults, ageing is often perceived as a significant hindrance to engaging in regular PA.[33 36–41 43 44] Pain associated with age and experienced during PA was a commonly cited obstacle,[25 27 31–33 36] with one study observing that over 95% of participants reported pain to be a limiting factor.[43] Additionally, participants expressed concerns about age-related decline in bodily function, such as balance,[25 35] endurance[27] and muscle capacity,[25–27 30–33 35 37 43] and minor surgeries,[30] as potential barriers to PA engagement. Moreover, in each of the three studies, more than half of the participants attributed their lack of PA engagement to poor health.[35 40 43]

Interest/motivation: Participants reported a lack of interest/motivation as a major barrier to PA engagement.[36 40 41] Common factors identified that could decrease an older adult's interest in participating in PA included boring PA sessions/instructors,[25] lack of knowledge about PA,[25 33] lack of interest in PA[36 40 41] and dislike of engaging in PA.[41]

Economic: Most studies reported that engaging in PA in gyms or through PA programmes was perceived to be too expensive for older adults.[27 30 31 39 44] The cost of transportation to access PA locations was also described to be unaffordable.[25 31]

Time restraints: Participants often reported not having enough time to engage in PA (reported by 45–82% of older adults),[31 33 36 37 39–41 44] that engaging in PA was too time consuming[42] or being too occupied with other tasks to engage in PA such as volunteer work.[35]

Interpersonal

Family and social support: Some older adults had full-time carer responsibilities which prevented them from engaging in PA.[30 39 44] Not having peer support[28 33–36 44] or friends to exercise with[36 37 40 43] also prevented people from engaging in PA. Females reported that they felt unsafe to engage in outdoor PA if they did not have a friend to exercise with.[43] Some participants' families were concerned about injury risks of exercise,[32 43] and some did not support the idea of the participant engaging in PA.[33 35]

Interaction with others: Older adults reported that engaging in PA with others was intimidating,[25] embarrassing,[26] while some were afraid of being rejected,[27] receiving third-party opinions and being judged.[36 43] One study reported that some participants perceived other participants' personalities to be a distraction, which negatively affected their mood to engage in PA.[28]

Language and cultural barriers: Another theme that emerged was language and cultural barriers faced by non-native/immigrant older adults when engaging in PA with others.[33] Older adults found it much more difficult to find a group to exercise with, when there were disparities to how PA was performed by different cultural groups.[44]

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Table 1 Summary of themes and subthemes influencing PA engagement within the socioecological model

Socioecological model	Theme	Barrier subtheme	Facilitator subtheme
Intrapersonal factors	Psychological	Fear of falling[25 27 30–37 39 40] Fear of fracture/damage to joints[26] Fear of getting hit by a car[34 43] Fear of getting assaulted[34] Fear of contracting COVID-19[29] No reassurance from doctors, osteopaths or physiotherapists[26 36 43]	Knowledge/awareness of PA benefits[25] Feel healthier after PA engagement[26 41] Improve body image[25 43] Reassurance from doctors, osteopaths or physiotherapists[36 43]
	Health-related	Effects of ageing[33 36–41 43 44] Pain[25 27 31–33 36 43] Loss of balance[25 35] Loss of endurance[25] Loss of muscle capacity[25–27 30–33 35 37 43] Surgery[30] Poor health[35 40 43]	Manage current/prevent future health conditions[25 26 33 39 41] Improve mental health[25 36] Maintain bodily movement and a sense of control[25–28 36] Wearable technology (smart watches and pedometers) and medical devices[33]
	Interest/motivation	Boring PA sessions/instructors[25] Lack of knowledge about PA[25 32] Not interested in PA[36 40 41] Dislike engaging in PA[41]	Enjoyable[25–28 33 34 36 42] Maintaining routine[25 26 29 32 36] Engaging PA instructors[28]
	Economic	Gyms or PA programmes are too expensive[27 30 31 39 44] High costs associated with using public transportation[27 31]	Publicly funded/subsidised gyms and PA programmes[30] Low cost activities[31]
	Time constraints	Not enough time to engage in PA[31 33 36 37 39–41 44] PA is time consuming[42] Occupied with other tasks[35]	Too much time,[30] Made time to engage in PA[39]
Interpersonal factors	Family and social support	Caring responsibilities[30 39 44] No peer support[28 33–36 44] No exercise partner[35 37 40 43] Family concerned about risks/injury[32 33] Family do not support PA engagement[33 35]	Family support[30 32] Support from trainers and perceiving them as friends[33] Support from friends[30 31 36] Had a friend to exercise with[41 42] Had groups of friends to exercise with[32 36] Being active not to become a burden to family[28] Pet companionship[25 33]
	Interaction with others	Intimidating[25] Embarrassing[26] Fear of rejection[27] Being judged[36 42] Personalities of others being a distraction[28]	Making new friends/socialising with friends[25 26 33 36 38] Compete with others[30 36] Compliment from others[33] Foster friendships[30]
	Language and culture	Cultural barriers[33] Different PA programmes due to culture[44]	Same language and background[33]
Environmental factors	Weather conditions	Poor weather[25 28 31–33 35–39 41 43 44] Hot or humid climates[35] Darkness[36 43]	Good weather condition[25 28 31–33 35–39 41 44]
	Access to PA locations/ facilities and built environments	No safe PA locations[25 27 29–36 39 40 42–44] Poor built environment[31 34 36] Park/PA locations are far away[25 33 34 36] No public transportation[34,39,40] Uneven walkways[25 27 34 36] No backyard/garden[29] Poorly lit streets at night[25 34] Little to no street signs/ security cameras[34] No community PA programmes[25 33]	Low cost/free PA facilities or programmes[25 26 30 31] Government funded PA programmes tailored to older adults[30] Well-lit streets[34] Available benches for rest[25 32 34] Aesthetically pleasing parks[34] Well maintained streets and wider sidewalks[36] Parks within walking distance[26 28 30 34 36] Had access to public transport[26 34 39]

PA, physical activity.

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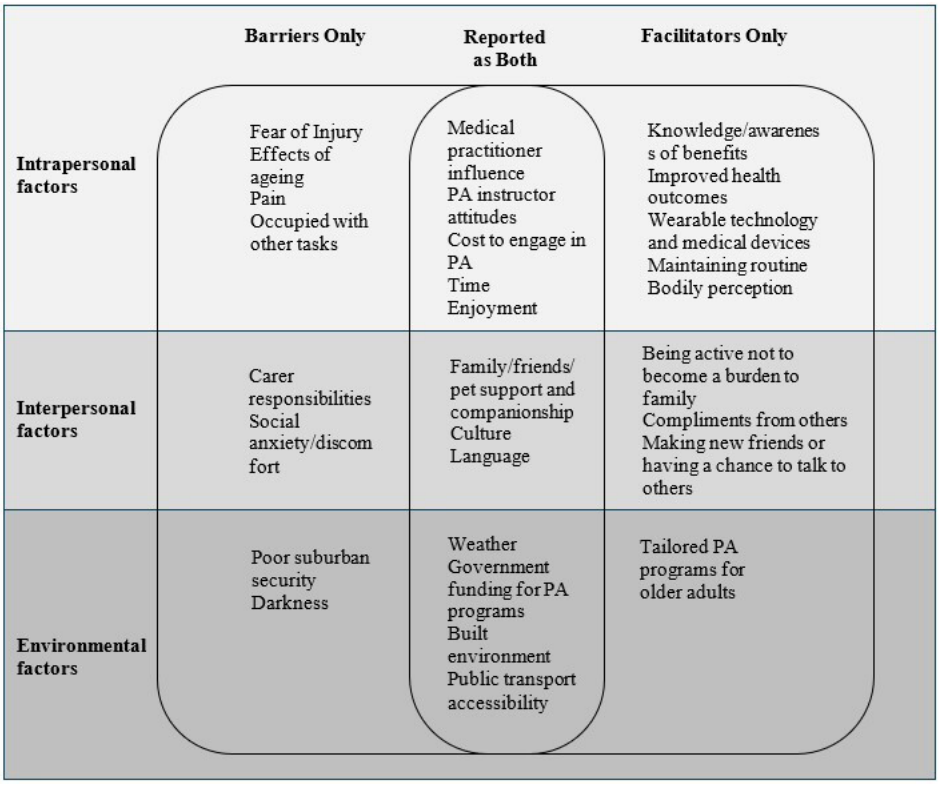


Figure 2 Venn diagram of subthemes categorised as only barriers, facilitators or both by each level of the socioecological model of health and behaviour. PA, physical activity.

Table 2 Summary of subthemes presented as both barriers and facilitators

Barrier	Facilitator
No reassurance from doctors, osteopaths or physiotherapists[26]	Having reassurance from doctors, osteopaths or physiotherapists[36]
Finding PA boring[25 33 35 36]	Enjoyable[25 28 33 42]
Boring PA instructors[25]	Engaging PA instructors[28]
Gyms or PA programmes are too expensive	Publicly funded/subsidised gyms and PA programmes
Not enough time[31 33 36 37 39 41 44]	Too much time[33]
Having no exercise partner[35 37 40 43]	Having a friend to exercise with[41 42]
Family do not support PA engagement[33 35]	Have family support[30 32]
Cultural barriers[33 44]	Same language and background[33]
Poor weather[25 28 31 33 35 39 41 43 44]	Good weather[25 33 39 41]
Poor built environments[31 34 36]	Well maintained streets and built environments[25 32 34 36]
No public transportation[34 39 40]	Had access to public transportation[26 34 39]
No community PA programmes[25 33]	Had community programmes tailored to older adults[30]

PA, physical activity.

Environmental

Weather conditions: Poor weather was the most commonly reported environmental barrier,[25 28 31–33 35–39 41 43 44] with one study from Australia reporting that 97% of participants found poor weather to be a deterrent to PA engagement.[39] Hotter or more humid climates were also reported to discourage PA engagement.[35] Darkness, particularly during winter in rural locations, further hindered PA engagement due to visibility concerns and safety risks.[36] It was revealed in a study from Finland that 89% of participants reported darkness to be a significant obstacle to PA engagement.[43]

Built environment and access to PA facilities and programmes: Older adults perceived barriers to PA engagement in the following built environmental factors: poor access to safe PA locations/facilities,[25 27 29–36 39 40 42–44] poorly lit streets or having little or no street signs and security cameras,[25 34] poor access to safe PA locations,[31 34 36] no walkways or poorly maintained sidewalks,[31 34 36] the nearest park or PA location too far away,[25 33 34 36] poor access to PA location due to a lack of or poor availability of transportation[27 34 39 40] or available PA locations may be of poor quality.[25 27 34 36] During the COVID-19 pandemic, it was also highlighted by older adults that they could not exercise during lockdown restrictions, as their houses did not have a garden/backyard.[29] Older adults commonly reported a lack of community PA programmes.[25 33]

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Facilitators of physical activity engagement

Intrapersonal

Psychological: Participants reported that by understanding the benefits of PA, they were more inclined to engage in PA.[25] Furthermore, some older adults reported that engaging in PA often made them feel healthier[26 41] and that engaging in PA could potentially improve their body image.[24 42] For those who were afraid to engage in PA, having reassurance from doctors, osteopaths or physiotherapists facilitated PA engagement.[36 43]

Health-related issues: It was commonly reported by older adults that engaging in PA could potentially assist them to manage their current or prevent future health conditions,[25 26 33 39 41] improve mental health,[25 36] maintain bodily movement coordination and a sense of control over their own body.[25–28 35] The application of technology and certain medical devices also helped older adults to be aware of the PA levels and notified them of when to go for a walk (watches and Fitbits will send notifications if the user is sitting for too long, help count daily steps, monitor sleep quality, warn the user if their heart rate is too high during PA engagement).[33]

Motivation: Older adults commonly reported that they engaged in PA because it was enjoyable.[25 28–33 42] For some, PA was already a part of their routine, and they would feel guilty if they did not engage in PA.[25 26 29 32 36] Others found their PA instructors to be quite engaging, which directly improved their motivation to continue to be physically active.[28]

Economic: It was reported that having access to publicly funded/subsidised gyms and PA programmes greatly improved PA accessibility.[30] Low base cost activities also enhanced PA engagement.[31]

Time restraints: For some older adults, life after retirement often left them with too much spare time and PA was a common method for them to pass time.[30] Others who enjoyed PA made time to engage in the PA that they enjoy.[39]

Interpersonal

Family and social support: Having support from family facilitated participants' engagement in PA.[30 32] Some older adults did not want to become a burden on their children and family, and engaging in PA was a way of avoiding becoming a burden.[33] Perceiving trainers as friends and ongoing support from trainers[28] and support from friends or groups of friends also facilitated PA engagement.[30–32 36 41 42] Some older adults also reported that having their pet accompany them greatly facilitated their engagement in PA.[25 33]

Interaction with others: For most of the studies, socialising with friends or making new connections was common themes that facilitated engagement in PA among older adults.[25 26 33 36 38] Some people perceived engagement in PA as a time to compete with others,[30 36] and they found pride and motivation when they were complimented by others.[33] It was also reported that by engaging in PA, women found a sense of community which allowed them to foster friendships.[30]

Language and cultural barriers: For older adults who were not fluent speakers in English, sharing a cultural background and language with others allowed them to access the same level of information and participate in PA programmes, which may have otherwise been difficult or inconvenient to obtain.[33]

Environmental

Weather conditions: Most participants reported that having good weather conditions made it easier to engage in PA.[25 29 31–33 35–41 44]

Access to PA programmes: Older adults reported that access to lower cost or potentially free PA programmes helped facilitate PA engagement.[26 26 30 31] Programmes that were engaging and tailored specifically for older adults made the PA classes more fun and less difficult to perform.[30]

Built environment: Several studies also reported that older adults preferred to engage in PA where community infrastructure was well maintained. This included having well-lit streets (especially at night),[34] benches to rest,[25 32 34] aesthetically pleasing parks,[34] well-maintained streets, wider sidewalks and quiet places to conduct PA.[36]

Having places within walking distance was also perceived as important. Older adults preferred to have stores, PA facilities or parks nearby.[26 28 30 34 36] If public transportation were available to take them to the PA locations, people were more likely to engage in PA.[26 34 39]

DISCUSSION

In this systematic review on the barriers to and facilitators of PA among community-dwelling older adults aged 65 or older, a content analysis was used to summarise the findings of the 20 included studies into themes and subthemes and categorised into three main levels: intrapersonal, interpersonal and environmental factors. *Intrapersonal* barriers included fear of injury, effects of ageing, pain and lack of motivation, while facilitators comprised health benefits, enjoyment and routine. *Interpersonal* barriers involved lack of social support and family obligations, whereas facilitators included support from family, friends and pets. *Environmental* barriers were poor weather, unsafe neighbourhoods and lack of accessible PA facilities, while facilitators included good weather, well-maintained environments and access to programmes. Many factors were reported to be both barriers and facilitators. These findings underscore the need for multilevel interventions targeting individual, social and environmental factors to enhance PA engagement among older adults.

Intrapersonal

Commonly reported intrapersonal barriers to PA engagement among older adults included fear of injury, pain, lack of motivation, limited spare time and the perception that safe PA locations were too expensive. These findings

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align with those of other systematic reviews that have focused solely on qualitative evidence among adults aged 60 and over,[15] or synthesised quantitative and qualitative evidence among frail populations or those with chronic conditions aged 80 or over.[16]

These findings suggest that personal attitudes, beliefs and perceptions significantly influence an individual's willingness to engage in PA. To increase PA uptake, it is essential to promote both the health and personal benefits of PA, while also addressing specific barriers experienced by diverse subgroups. For older adults living with or at risk of chronic disease, emphasising the protective effects of PA may help facilitate greater engagement. This is supported by the health belief model, which posits that perceived health threats can serve as cues to action, and that perceiving benefits from a behaviour can support its adoption and maintenance.[45] This may also explain why in this review most older adults reported being motivated by improvements in health and well-being, enhanced body image, a greater sense of control over their bodies and the enjoyment they derived from. For those with cognitive impairment, such as cognitive decline or dementia, caregivers may play a critical role in supporting PA participation—by providing reminders and encouragement—while reinforcing the message that any movement counts.

Interpersonal

A lack of support from peers or family, along with family obligations, emerged as key interpersonal barriers to PA engagement among older adults. This highlights the central role that family and friends play in shaping older adults' health behaviours and overall well-being. General practitioners and PA programme providers should therefore promote PA not only to older adults, but also to their families and social networks, encouraging family and community inclusiveness. Community-based PA programmes should emphasise enjoyment and social connection—through activities such as dancing or golfing—rather than framing PA solely as means of preventing ageing and improving health.

Support from gym instructors and peers who also engage in PA was also found to play a crucial role in facilitating PA participation. A systematic review on social engagement and PA in older adults reported that higher levels of social support were associated with a greater participation in leisure-time PA and significantly reduced self-reported loneliness.[46] This highlights the importance of fostering socially supportive environments that extends beyond immediate family. However, whether social support can enhance engagement in moderate or vigorous intensity PA remains unclear and warrants further investigation.[46]

Medical technologies have been reported to facilitate PA engagement; however, their effective use is highly dependent on digital health literacy and cost.[47 48] Previous research has highlighted that poor health literacy can limit the use of such technologies; in particular, older adults may be less likely to use technology to support PA if they are not confident or familiar with how to use it.[47] Therefore, it is important for practitioners to support and educate older adults in using medical technologies during consultations, to help maximise their potential benefits.[47] Similarly, the cost of technology, particularly smart technology, can significantly limit one's ability to use fitness apps, online PA programmes and pedometers.[48] A systematic review of digital interventions for increasing PA among low socioeconomic status (SES) individuals reported that digital access to PA interventions is not equitable between low and high SES groups.[48] Moreover, there is little evidence demonstrating efficacy in health outcomes for low SES individuals who do use digital technologies for PA.[48] The study also acknowledges the difficulty in grouping low SES populations due to varying classifications.[48] Therefore, establishing universal classifications of SES is essential to enable a more comprehensive exploration of the relationship between digital access and PA engagement.

Environmental

Poor weather was the most reported environmental factor affecting older adults' participation in PA. This highlights the importance of adapting PA programmes to the changes in weather conditions. For example, offering indoor community-based or online PA sessions during periods of inclement weather can help maintain regular engagement.[44] Additionally, indoor programmes scheduled in the evening could support those who may feel unsafe being active outdoors at night.[44]

Older adults frequently reported limited access and the high cost of PA facilities as major barriers to PA engagement. These findings are consistent with an earlier systematic review of qualitative evidence, which suggested that individuals with low SES often face restricted access to PA programmes and locations.[15] Beyond constructing new facilities, local governments should consider improving public transportation options to existing PA locations and reducing transport costs for seniors to enhance equitable access.[15] Another potential solution is to integrate PA programmes into existing health coverage benefits under reciprocal healthcare agreements (RHCA), such as Medicare in Australia, which becomes available at age 65. One example is the SilverSneakers (SS) programme in the USA.[49] In a cost-effectiveness study by Nguyen *et al*, involving 17 801 participants enrolled in SS programmes in 2007, those who participated at least once per week for 2 years reduced their annual healthcare costs by approximately US$500 compared with non-participants.[49] Those engaging two to three times per week saved around US$1200 annually.[49] These savings were largely driven by fewer and less costly inpatient visits.[49] For countries that already include PA under RHCA, strategies to raise awareness and promote consistent PA programme participation are essential. For instance, although the SS programme is effective, healthcare providers in the US may mention it during biennial preventive health visits, but no structured follow-up processes ensure sustained engagement by older adults.[49]

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Implication of study results for health policy and future research

As people age, their ability to engage in PA may change, and some may not be able to meet recommended guidelines.[3] Therefore, current practice should promote the message that *any movement counts*, emphasising PA options that align with individuals' physical capabilities. This includes activities such as walking programmes, aqua aerobics, yoga/mobility classes and condition-specific classes. For those unable to participate in structured PA programmes, everyday activities like walking for transport, gardening or domestic chores should also be encouraged. Adopting a person-centred approach can help reduce fear of injury associated with exceeding one's physical capacity.[10] Another important consideration is shifting a prevailing mindset around PA—from one that views it primarily as a preventive or health promoting behaviour to one that emphasises enjoyment and pleasure.[33] Governments should also work toward improving the accessibility and connectivity of public transportation to PA locations and consider implementing national funding programmes that incentivise PA participation among older adults.[15 48]

The findings of our review summarise the intrapersonal, interpersonal and environmental barriers to and facilitators of PA engagement in healthy community-dwelling older adults. Future research should investigate these factors among specific subgroups, such as individuals with differing education levels or low SES for each country, so that targeted strategies can be implemented to the local social, environment and cultural context.[14]

Strengths and limitations

Our study is among the first to synthesise qualitative, quantitative and mixed-methods evidence on the barriers to and facilitators of PA among community-dwelling older adults aged 65 years or older. It extends the evidence base, which has previously focused on qualitative research in adults aged 60 years and over, or on evidence from those aged 80+ years living with frailty or chronic conditions,[16] or from adolescent populations.[50] Focusing specifically on community-dwelling adults aged 65+ is critical, as this group represents a large, growing and relatively independent segment of the older population. Understanding the unique factors that influence PA in this population who are not necessarily frail or institutionalised is essential for designing effective, scalable and preventative interventions. Relying solely on evidence from more vulnerable groups may lead to overmedicalised or less acceptable approaches that fail to engage healthier older adults and miss opportunities for early, sustainable behaviour change. Additionally, rigorous synthesis of data was conducted using the socio-ecological model of health behaviour to frame themes and subthemes into categories of intrapersonal, interpersonal and environmental factors.

Despite its strengths, this systematic review has several limitations. Most included studies were conducted in high-income countries with predominantly Caucasian participants, limiting the generalisability of the findings for older adults living in low- and middle-income settings or from other ethnic backgrounds.[33] Additionally, restricting inclusion to English language publications may have excluded relevant evidence published in other languages. Finally, the barriers and facilitators identified in this review may not reflect the experiences of older adults residing in care facilities, particularly those living with functional disabilities or life-limiting conditions.[13]

CONCLUSION

The results of our study provide a more comprehensive understanding of the multifaceted factors influencing PA engagement among community-dwelling adults aged 65 and over who are not functionally disabled or receiving palliative care. While older adults generally recognise the benefits of PA for healthy ageing, a range of intrapersonal factors (eg, fear of injury, pain, low motivation and lack of spare time), interpersonal factors (eg, lack of social support and family obligations) and environmental factors (eg, poor weather, lack of access/high cost accessing PA facilities) were identified as a barrier to PA engagement. To enhance PA participation at the population level, governments should consider multilevel strategies that target individual behaviours, community engagement and supportive health policies. Efforts should also harness key facilitators to PA, including positive intrapersonal motivators (eg, improved health and well-being, body image, sense of control and enjoyment), interpersonal enablers (eg, social support, exercising with partners or groups) and environmental support (eg, age-friendly infrastructure and government-subsidised programmes). Policymakers and clinicians are encouraged to shift the prevailing narrative of PA promotion from a solely health-focused approach to one that emphasises the enjoyment, social connection and inclusiveness of PA through diverse, accessible activities suitable for all abilities.

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